

Luiz Sérgio da Silva Filho

SOFTWARE ENGINEER | FULL STACK DEVELOPER

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Open to hybrid, remote or on-site roles

SUMMARY

Full stack software engineer with ten systems of my own delivered end to end, from architecture to deployment, across domains such as labor-court expert work, school management, automotive marketplaces and personal finance, one of them already running in production with real customers. The edge is in the rigor behind delivery: money as Decimal and never a float, balances derived from an append-only ledger, multi-tenant isolation with Row-Level Security verified by explicit tests, idempotent schedulers and webhooks, and AI layers that never write to the database without application-side validation. Bachelor's in Information Technology, with a background in IT support, running my own business and pedagogical coordination, experience that keeps technical decisions close to the real pain of the people using the system.

PROFESSIONAL EXPERIENCE

Pedagogical Coordinator

Feb 2026 – Present

Children's Programming and Robotics School · Barretos, SP, Brazil

- Built the pedagogical coordination from the ground up when the course opened: organized the classes, defined the curriculum by age group and assembled the teaching team, serving students from 7 to 18 years old.
- Age-progressive curriculum: younger classes with robotics (Arduino, micro:bit) and block-based programming; teenagers starting with Python and moving on to Git, HTML, CSS, JavaScript, React, Node.js, MongoDB and JWT as their full stack foundation.
- Teach Python to the teenage classes, covering programming logic, basic data structures and small hands-on projects.
- Faced with the limitations of the system the unit was using for pedagogical management and tracking, I ran the entire lifecycle of a replacement system on my own: gathered the requirements from my own coordination routine, designed the data model and the role-based permissions (director, coordinator, teacher), and validated every release against real school usage before putting it into production. The product is Atlas Educação, detailed below.

Owner and Partner

Jun 2020 – Present

Empório do Bebê · Barretos, SP, Brazil

- Responsible for running the business: point of sale and inventory control, online sales channel and digital catalog, and automation of administrative processes through spreadsheets. The systems built in parallel are detailed in the next section.

IT Support Analyst

Aug 2013 – Dec 2019

Sesi São Paulo · Limeira and Barretos, SP, Brazil

- IT support for staff, teachers and students: helpdesk ticket handling, administrative and educational systems, and support for classroom technology.
- Technical mentoring for educational robotics teams: support with programming and with planning the execution order of the robot missions to solve the proposed challenges, in competitions such as FLL (First Lego League) and OBR (Brazilian Robotics Olympiad).

INDEPENDENT TECHNICAL PROJECTS

Complete systems conceived, architected and shipped independently, alongside my work as a partner at Empório do Bebê. Technical breakdown of every project at luizsergio.vercel.app.

- **Sondei** · sondei.org · Automotive marketplace in a monorepo, with the full FIPE price table (11,354 versions) and IBGE geography imported, backed by 443 integration tests against a real Postgres. The recommendation advisor never gets a database connection: it only extracts filters from the question and answers on top of an application-controlled query, making it impossible to suggest a vehicle that is not in stock. Turborepo, Next.js, Drizzle ORM.

- **Tonico Veículos** · tonicoveiculos.com.br · Platform delivered to a pickup-truck and used-car dealership, live and in daily operation. Monetary values as integer cents, accent-insensitive search solved in the application layer because the managed Postgres lacks the extension, and photos carrying their own dimensions because of the legacy catalog. Next.js 16, Prisma, JWT.
- **Liquida Sorte** · liquidaemporium.sa2labs.com.br · In production, serving real customers. Inventory liquidation in lots and cards, with auditable draws and cryptographically generated random numbers. Customer balance is always derived from an append-only ledger, never edited directly, and every payment webhook is idempotent. Checkout and draws sit behind feature flags that are off by default.
- **Atlas Educação** · atlas-educacao.vercel.app · In use at the school where I work as coordinator. Multi-school portal with a dropout-risk engine whose attendance, absence and grade-drop thresholds are configurable per school. A seven-provider LLM chain, personal data anonymized before anything is sent, and a product that stays fully functional with no AI key.
- **Meu Assessor** · acessor.sa2labs.com.br · Personal finance over WhatsApp and a web dashboard, with a domain core in plain TypeScript following DDD. Money always as Decimal and never a float, exact cent distribution across installments and schedulers idempotent by their own key. Argon2id, TOTP-based MFA and data-protection routines. NestJS, Prisma, BullMQ.
- **PeritoCalc** · SaaS for court-appointed labor experts. The extraction engine routes every field through eight cost-ordered layers, from native PDF text to computer vision as a last resort, with a per-layer auditable ledger. The .PJC file generator runs in pure Python 3 with zero external dependencies and produces the XML ready for import into PJe-Calc.
- **Other systems:** Gnatus Verificação, which checks document authenticity server side instead of trusting the AI reading; multi-site time tracking with facial recognition in an isolated Python microservice; retail ERP in NestJS and Fastify with nine-role RBAC and a pluggable tax layer; and Canal Dark, an Electron desktop app for video production with a bundled FFmpeg.

EDUCATION

Postgraduate Degree in Higher Education Teaching <i>Faculdade XV de Agosto</i>	2015
Bachelor's Degree in Information Technology <i>Faculdade Barretos</i>	2008 – 2011

TECHNICAL SKILLS

Frontend & languages: TypeScript, Python, React.js, Next.js
Backend & APIs: Node.js, NestJS, Fastify, REST API, Webhooks
Data & architecture: PostgreSQL, Prisma, Drizzle, Row-Level Security, Multi-tenant, Redis, BullMQ
Artificial intelligence: Claude API, OpenAI API, structured extraction, tool-calling, guardrails
Engineering & infrastructure: Docker, CI/CD, Git, integration testing, Stripe, n8n